

Harshana Lakshara Fernando

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Summary

aBackend-focused Computer Science Engineering undergraduate with a rigorous foundation in **System Design**, **Distributed Systems**, and **Cloud-Native Engineering**. Proven experience architecting high-performance, fault-tolerant applications using **Microservices**, **Serverless Architectures**, and **Event-Driven Design** with **Java (Spring Boot)**, **Go**, **AWS Lambda**, and **Kafka**. Validated through the delivery of high-throughput systems handling massive load, seeking to solve complex **concurrency** and **low-latency** challenges within high-velocity engineering teams.

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering

2022 – 2026

University of Moratuwa, Sri Lanka

G.C.E. Advanced Level (Physical Science Stream)

2019

St. John's College, Panadura, Sri Lanka

Results: 3 A's (Combined Mathematics, Physics, Chemistry)

District Rank: 17

Experience

Research

Software Engineer Intern

Mar 2026 – Present

Empathic Computing Laboratory, University of Auckland (Top 100 QS, Remote)

- **KōreroCare: Secure Mobile–Wearable Digital Phenotyping Infrastructure**

- Designed a research-grade digital phenotyping infrastructure for longitudinal behavioral and physiological data collection across Android, WearOS, and backend services.
- Engineered reliable passive sensing workflows using Foreground Services and WorkManager to sustain high-uptime data capture under Android OS constraints.
- Built resilient synchronization, local buffering, and retry mechanisms to reduce data loss and improve consistency across mobile and wearable data streams.
- Contributed to backend architecture using Go, with service-oriented design for scalable ingestion and secure processing of high-volume time-series data.
- Integrated Keycloak for authentication and identity management, and introduced Traefik as an API gateway for centralized routing in a containerized environment.
- Explored Open Wearables integration to support interoperable ingestion and normalization of wearable data across heterogeneous device ecosystems.

KEYWORDS: Kotlin, Android, WearOS, Go, Keycloak, Traefik, Open Wearables, FastAPI, Docker, WorkManager **Software Engineer Intern (Fullstack Focus)**

Dec 2024 – May 2025

Information Systems Associates (Pvt) Ltd — Aviation Technology Company

- Engineered a distributed, high-throughput unique ID generator for Passenger Name Records (PNRs) using a Kafka-backed pool model, ensuring uniqueness and security across scaled instances while adhering to aviation domain standards.
- Architected and implemented a hybrid integration strategy by embedding a gRPC client and Kafka consumer into a legacy Java 8 monolith (AeroMart), enabling seamless, backward-compatible communication with the new AeroOrder microservices platform.
- Developed dynamic, database-driven feature toggles to control PNR routing logic, minimizing risk during the phased production rollout by allowing for granular control and immediate fallback.
- Actively contributed to the agile development lifecycle, participating in all ceremonies, performing code reviews, and collaborating with cross-functional teams to deliver production-ready features using Jira and Bitbucket.
- Integrated frontend components with backend services to deliver end-to-end features.

Keywords: Java 18, Spring Boot, React, gRPC, Kafka, REST API, PostgreSQL, Docker, Microservices, Agile.

Projects

See more projects on my portfolio 🔗

1. Employee Management System (EMS) (2025 | Individual) Architecture | Github

Designed and built a comprehensive, cloud-native microservices application for employee and department management, featuring a full-stack architecture with a React/TypeScript frontend and Spring Boot backend.

- Architected a distributed system using **Spring Cloud** for service discovery, centralized configuration, and an API gateway, with **Resilience4j** for robust fault tolerance.
- Enabled dynamic configuration management using **Spring Cloud Bus** with **RabbitMQ** and established a robust observability stack (Zipkin, ELK, Prometheus/Grafana).
- Designed and managed a CI/CD pipeline using **GitHub Actions**, with **ArgoCD** for GitOps-based deployments to the Kubernetes cluster.

Keywords: Java 17, Spring Boot, Spring Cloud, React, TypeScript, Resilience4j, RabbitMQ, Docker, Kubernetes, GitHub Actions, ArgoCD, Prometheus, Grafana

2. GoTogether (2024 | Group) Architecture | Live | Github

Developed and deployed a full-stack travel assistant platform with a **Next.js** frontend (SSR/CSR, secure auth) and microservices in **Spring Boot** and **Go**, enabling AI itinerary generation via **Gemini API**.

- Implemented a **Spring Boot** social service with **Keycloak** integration for identity management and a dedicated **Go** service using **gRPC** for high-performance inter-service communication.
- Configured production-grade security with **NGINX**, **Certbot**, and **DuckDNS**, and orchestrated deployment on **Oracle Cloud Free Tier** using custom scripts and Docker optimizations.
- Built a dedicated **Auth Service** using **Keycloak**, managing OAuth2/OIDC flows, token issuance, and automated realm/client boot-time import scripts.

Keywords: Next.js, Go, Spring Boot, Keycloak, PostgreSQL, NGINX, Docker, Gemini API, Oracle Cloud, gRPC

3. Food Delivery Application (2024 | Group) Github

Built a scalable, event-driven food delivery platform inspired by systems like Uber Eats, using a modern microservices architecture. As part of the core team, I focused on designing and developing the Restaurant Service.

- Developed the Restaurant Module with secure sign-up/login using **Spring Security** and **JWT** (with extended claims).
- Built CRUD APIs for menu and item management; integrated **Redis caching** to improve performance and reduce database load, with appropriate cache invalidation strategies.
- Enabled order sync via **Kafka**: when an order is placed, an event is published to the 'order_created' topic, which the restaurant service consumes to update order details.

Keywords: Java, Spring Boot, gRPC, Kafka, Redis, PostgreSQL, WebSocket, Docker, React.js, Ant Design

4. Maison Vella – E-Commerce Admin Panel (2024 | Individual) Architecture | Live | Github

Comprehensive admin interface for the Maison Vella e-commerce ecosystem, facilitating centralized management of products, orders, and content assets.

- Engineered a responsive **Next.js 14** app with **Server Actions**, **tRPC**, and **ShadCN UI**, implementing **CQRS** and layered architecture.
- Integrated **Sanity** (Headless CMS) via GraphQL and managed data with **Prisma** and **Supabase** in a serverless environment.
- Secured access with **Clerk Auth** and optimized media delivery using **Cloudflare R2** and a global CDN.

Keywords: Next.js 14, Server Actions, tRPC, GraphQL, Prisma, Supabase, Sanity, Serverless, Cloudflare R2, CQRS

5. Virtual Me – 3D Animated AI Chatbot (2024 | Individual)

Created an interactive 3D virtual assistant with lifelike animations and AI-driven conversations.

- Integrated **Hugging Face** (Mixtral model) for natural language dialogue.
- Implemented **Eleven Labs** TTS for human-like voice output.
- Built immersive 3D interface with **React** and **Three.js**.

Keywords: React (Three.js), Node.js, Express.js, Hugging Face, Eleven Labs, Rhubarb, FFmpeg

Professional Training, Coursework & Certificates

[See all certificates !\[\]\(a73c1962d20a39dd8fd6a060ae69693f_img.jpg\)](#)

• Certified Kubernetes Administrator (CKA) Training

KodeKloud (Mumshad Mannambeth) — Completed 27h curriculum with 50+ hands-on labs.

Apr 2025

[View certificate](#)

• AWS Certified Solutions Architect Associate (SAA-C03) Training

Udemy (Stephane Maarek) — Completed 27.5h comprehensive cloud architecture course.

Aug 2025

[View certificate](#)

- Apache Kafka Series (v3) – Learn Kafka for Beginners**
Udemy (Stephane Maarek) — Full System Design & Setup. *Mar 2025*
[Part 1] | [Part 2] | [Part 3]
- 100 Days of DevOps (Bootcamp)**
KodeKloud — Intensive hands-on training covering Linux, Python, Docker, Jenkins, Terraform & Ansible. *Nov 2025*
View certificate
- Implementing Software Architecture for Large-Scale Systems**
Udemy *Apr 2025*
View certificate
- Advanced OAuth Security**
Udemy *Mar 2024*
View certificate
- Next.js: Build Scalable React Apps with Page & App Routers**
Udemy — Instructors: Anton Voroniuk, Denys Kohut, Anton Voroniuk Support *Nov 2025*
View certificate

Skills

Core	Java Spring Boot Microservices Distributed Systems System Design REST APIs
Backend	Spring Cloud Spring Security Resilience4j Node.js Express.js Gin
Messaging & APIs	Kafka RabbitMQ gRPC GraphQL WebSockets Protobuf JWT OAuth 2.0
Data	PostgreSQL MySQL Oracle DB Redis Couchbase
Cloud & DevOps	AWS Docker Kubernetes CI/CD (GitHub Actions, Jenkins)
Testing	JMeter Gatling Pytest Unit Testing
Tools	Git Gradle Postman Jira Confluence

References available upon request.